

Self-Study Lecture no 1

In Homework Solutions no 3, we have seen that the model for the experiment described in Problem 3.31 is a one way random effects ANOVA:

$$Y_{ij} = \mu + \tau_i + \varepsilon_{ij}, i=1,2,3,4 \text{ and for a specified } i, j=1,2,3.$$

From the SAS output no1 for Problem 3.31, discussed in Homework Solutions no 3, it was evident that the variances of the random errors are increasing. It was also noted that the plot of the jackknifed residuals vs predicted values shows that there is one mild outlier. Therefore, one has two options: to remove the outlier or do a transformation, and then reanalyze the data.

First, let us remove the outlier and reanalyze the data. This is done in SAS file no 2 for Problem 3.31 and the results are given in SAS output no 2 for Problem 3.31. As we have seen before, by removing an observation in a data record and replacing it with a period we are telling SAS that we have a missing observation. From the SAS output no 2 for Problem 3.31, we see that the plot of Residual vs Predicted Value has no patterns and the plot of jackknifed residuals vs predicted values has no outliers. The q-q (residual vs quantile) plot looks fine and the Cook's D values are less than 1. The p-value of the W statistic is about .46.
The removal of the outlier worked!

a. Since this is a random effects model we are testing:

$$H_0: \sigma^2_{\tau} = 0 \text{ vs. } H_a: \sigma^2_{\tau} > 0.$$

Based on SAS output no 2, page 2, in the Type III SS table, in the attached SAS output, the value of the F statistic for testing this null hypothesis is 28.08 and the corresponding p-value is .0003.

Therefore, we reject the null hypothesis at the .05 level and conclude that there is significant variation in the uniformity between wafer positions.

The output of the PROC VARCOMP gives us the estimates for the variance components.

b. The variability due to wafer positions is estimated by:

$$\hat{\sigma}^2_{\tau} = 2.31$$

and

$$\hat{\sigma}^2_{\varepsilon} = \text{MSE} = 0.23$$

Therefore, $\hat{\sigma}^2_{\text{Total}} = 2.31 + 0.23 = 2.54$.

The proportion of variability due to wafer position is $\hat{\rho} = 2.31/2.54 = .9094$. Which means that about 91% of the variability is due to wafer position.

Now let us try the transformation approach. Let us transform the response variable using the LOG transformation. This is done in SAS file no 3 for Problem 3.31 and the results are given in SAS output no 3 for Problem 3.31.

From the SAS output no 3 for Problem 3.31, we see that the plot of Residual vs Predicted Value has no patterns and the plot of jackknifed residuals vs predicted values has no outliers. The q-q (residual vs quantile) plot looks fine and the Cook's D values are less than 1. The p-value of the W statistic is about .41.

The transformation of the data has worked as well!

Based on SAS output no 3, page 2, in the Type III SS table, in the attached SAS output, the value of the F statistic for testing this null hypothesis is 9.33 and the corresponding p-value is .0054.

Therefore, we reject the null hypothesis at the .05 level and conclude that there is significant variation in the uniformity between wafer positions.

The output of the PROC VARCOMP gives us the estimates for the variance components.

b. The variability due to wafer positions is estimated by:

$$\hat{\sigma}^2_{\tau} = 0.7415$$

and

$$\hat{\sigma}^2_e = \text{MSE} = 0.0795$$

Therefore, $\hat{\sigma}^2_{\text{Total}} = 0.7415 + 0.0795 = 0.821$.

The proportion of variability due to wafer position is $\hat{\rho} = 0.7417 / 0.821 = 0.9032$

Which means that about 90% of the variability is due to wafer position.

To summarize, the conclusions for the data analysis using these two approaches is the same. The advantage of using the first approach, by removing the outlier is that we have analyzed the original data (minus the outlier), rather than the transformed data.

Problem 4.14. An article in *Communications of ACM* Vol. 30, No. 5 1987, investigated different algorithms for estimating software development costs. Six algorithms were applied to six different software development projects, selected by the experimenter, in a completely randomized fashion. The percent error in estimating the development cost was recorded. The data set is given in our textbook on page P-18.

Solution.

4.14. a. Based on the description of the experiment, this is a completely randomized block design. The treatment factor is the algorithm, which has 6 levels; and the block factor is the software development project, which has six levels. Both factors are fixed. The response variable is percent error in estimating development cost. The model for this experiment is:

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, i=1, \dots, 6, j=1, \dots, 6. \quad (1)$$

In this experiment we are interested in testing the following null hypothesis:

$$H_0: \tau_i = 0, i=1, \dots, 6.$$

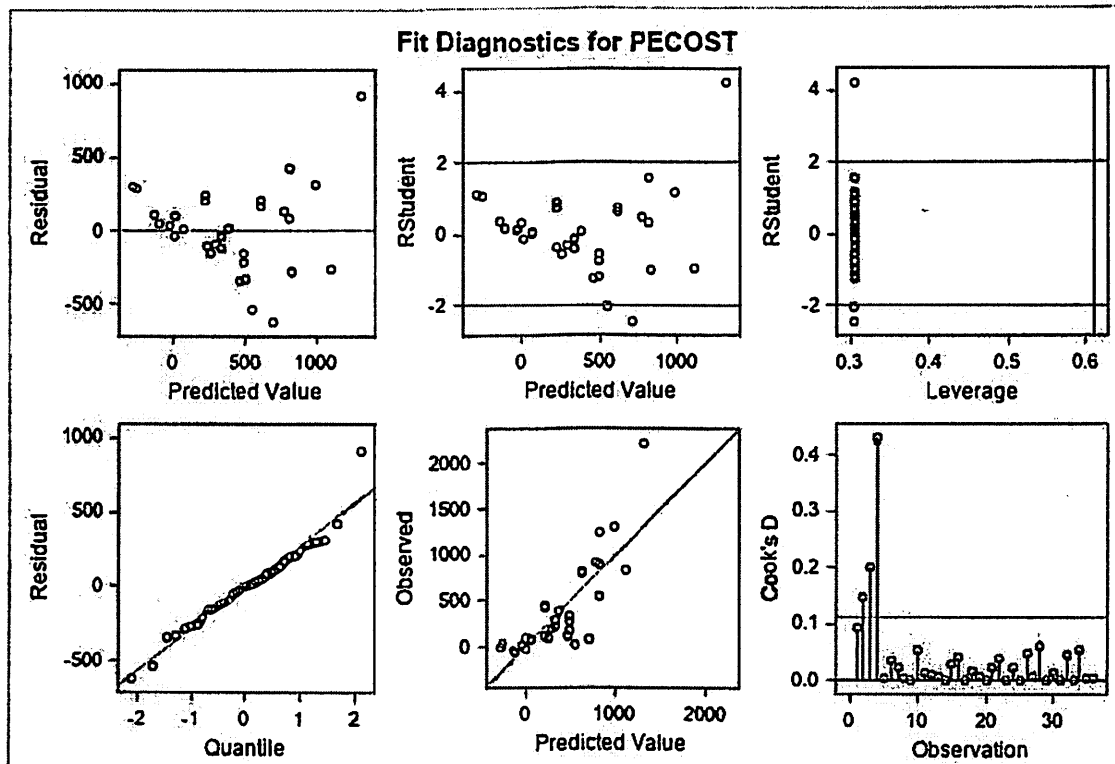
Assuming that all the underlying assumptions for this model are valid, based on the following Type III SS table:

Source	DF	Type III SS	Mean Square	F Value	Pr > F
ALGORITHM	5	2989130.472	597826.094	5.38	0.0017
PROJECT	5	2287339.472	457467.894	4.11	0.0073

We can reject the null hypothesis at the $\alpha = .002$ level and conclude that the levels of the algorithms have a significant effect on the percent error in estimating development cost.

This conclusion is valid if all the underlying assumptions for the model are valid. If that is true, then one proceeds to Tukey multiple comparisons of the means of percent error in estimating cost for the 6 algorithms.

b. To examine the validity of the underlying assumption for model (1), we examine the Fit Diagnostics plots from the SAS output presented below:



First, we note that there is pattern in the plot of the Residual vs. Predicted value: an increasing triangular or megaphone like pattern. This suggests that the variances of the random errors are increasing. In the plot of the RStudent vs. Predicted values one observation appears to be a potential mild outlier as its value exceeds 4, but not by much. The normal probability plot is pretty straight, except for one point, and the p-value of the W statistic, based on the SAS output for model (1), is equal to about $.24 > .10$.

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.961395	Pr < W	0.2373

Therefore, the normality assumption for the random errors is valid.

Since the variances of the random errors are increasing, one has to transform the data to stabilize the variances, and reanalyze the data. Since we have negative percentages, and a log transformation might be the first one to try, we will add 100 to the data points and then apply the log transformation. The following model is considered:

$$\log(Y_{ij} + 100) = \mu + \tau_i + \beta_j + \varepsilon_{ij}, i=1,\dots,6, j=1,\dots,6. \quad (2)$$

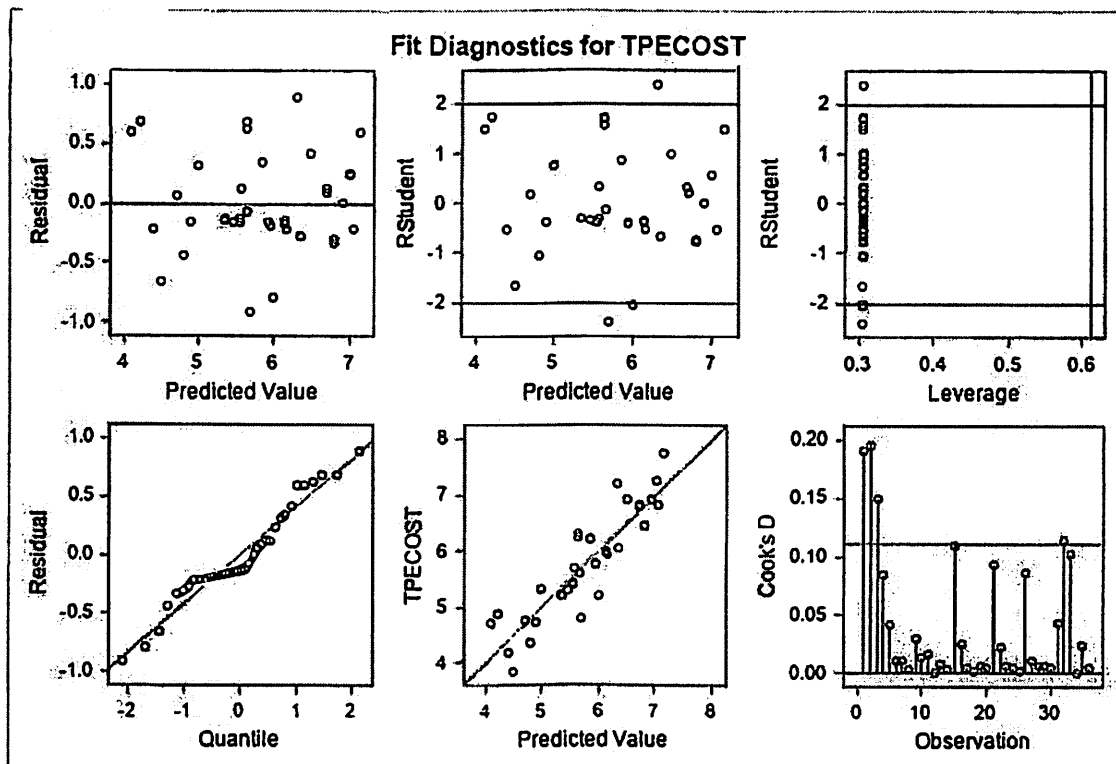
We are interested in testing the following null hypothesis:

$$H_0: \tau_i = 0, i=1,\dots,6.$$

Based on the SAS output for model (2) we get the following Type III SS table:

Source	DF	Type III SS	Mean Square	F Value	Pr > F
ALGORITHM	5	14.68532518	2.93706504	12.05	<.0001
PROJECT	5	9.86759742	1.97351948	8.10	0.0001

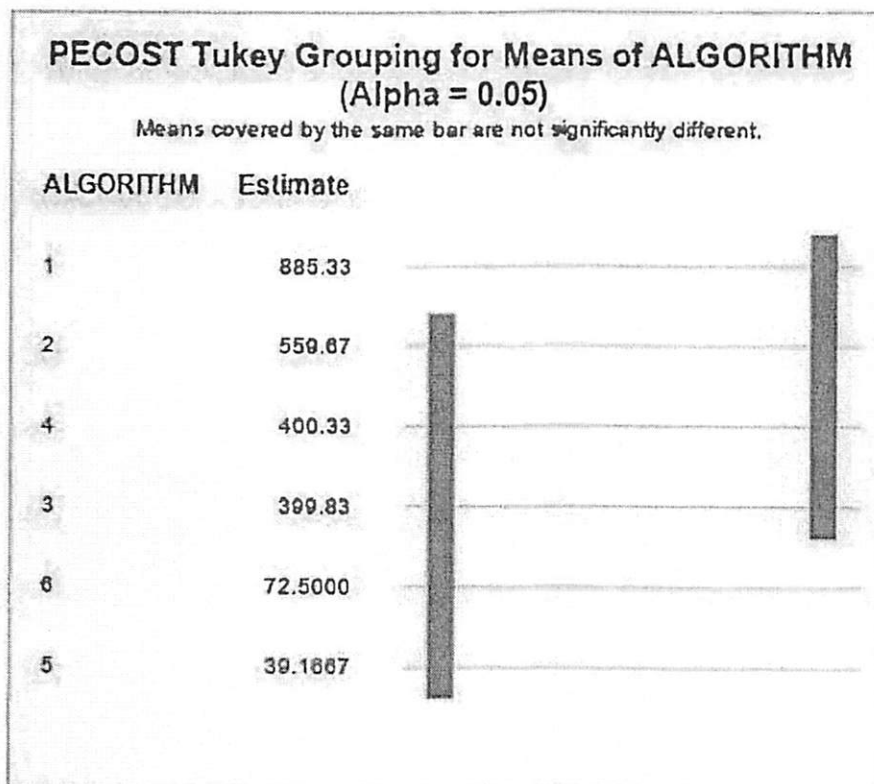
Since the p-value of the F statistic is less than .0001, we reject the null hypothesis stated above and conclude that the algorithms have an effect on the error in estimation development cost. This conclusion is valid if the underlying assumptions for model (2) are valid.



Note that the transformation we have applied has worked! There are no patterns in the plot of the Residual vs. Predicted value. RStudent values are all considerably less than 4 and greater than -4. The normal probability plot is pretty straight and the p-value of the W statistic for model (2) is: .12 > .10.

Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.951893	Pr < W	0.1200

c. The following graphical display is given in the SAS output for model (1):

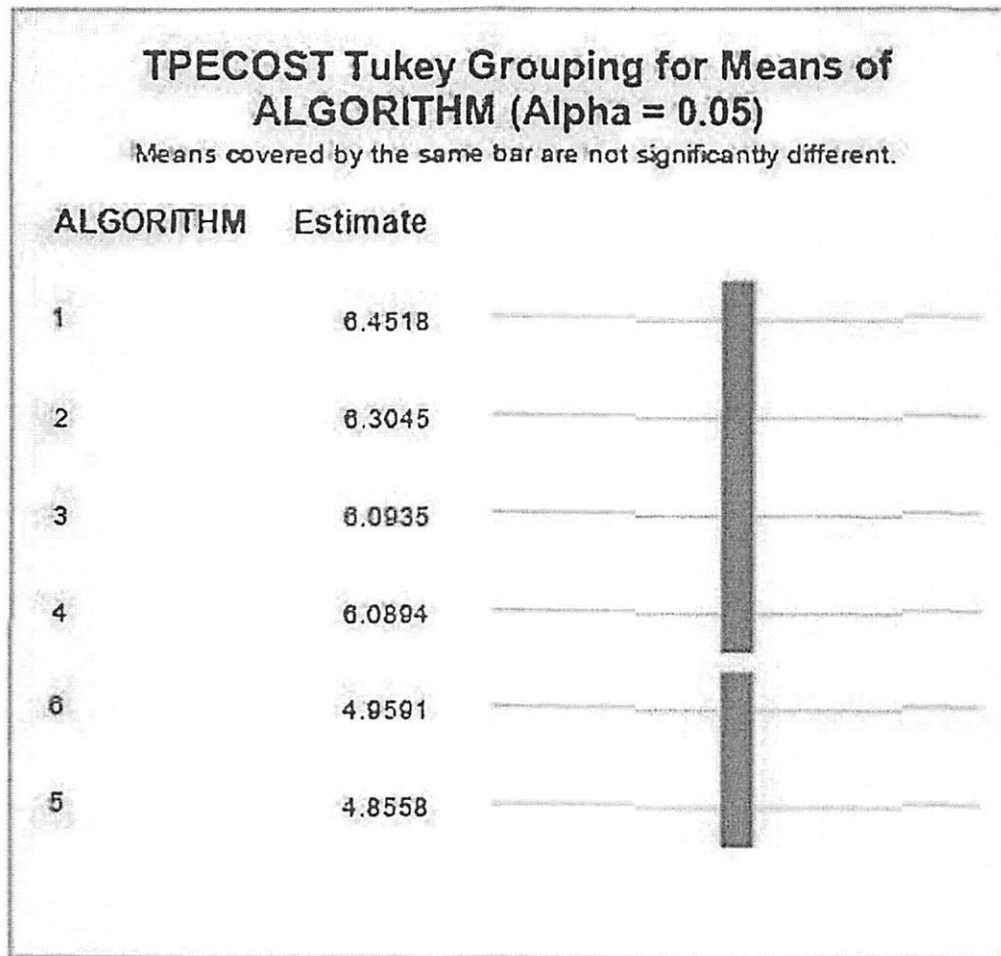


Based on the Tukey multiple comparisons for model (1) we can conclude that Algorithm 1 has a significantly higher mean percent error in estimating development cost than Algorithm 5.

Algorithms 1,2, 4 and 3 are not significantly different. Also, Algorithms 2,4,3,6 and 5 are not significantly different.

Note that the validity of the above Tukey multiple comparisons is questionable as it is also based on the validity of the underlying assumption for model (1).

From the SAS output for model (2) we get:



Based on this plot we can conclude that algorithms 5 or 6 should be recommended for use in practice, as they have a significantly lower mean percent error in estimating cost. Note that one has to transform the estimates in the display above to the original units. For algorithms 5 and 6, the estimates in the original units are:

$$\exp(4.8558) - 100 = 28.5\%$$

$$\exp(4.9591) - 100 = 42.5\%$$